

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think>
Total size of the file = 200 GB
Initial download rate = 2 GB / minute
Time taken to download 200 GB = $200 / 2 = 100$ minutes
Since 40 % of the way through the download , the download is interrupted and restarted , Carla has to download 40 % of the file again , which is $0.4 * 200 = 80$ GB
Time taken to download 80 GB = $80 / 2 = 40$ minutes
Total time taken for the download to be interrupted and restarted = 20 minutes
Total time taken to download the entire file = $100 + 40 + 20 = 160$ minutes
</think><answer> 160 </answer>

